

"F O R M S" - Creating new visual perceptions of dance movement through machine learning

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ABSTRACT

"FORMS" is a new digital art concept that combines the fields of dance movement and machine learning techniques, specifically human pose detection, to create a real-time and interactive visual experience. This project aims to explore the relationship between dance and visual art by creating a framework that generates abstract and literal visual models from the dancers' movements. The main objective of this project is to enhance the perception of dance movement by providing a new layer of visual composition. The proposed framework provides different visual forms based on human pose detection, creating a novel and real-time visual expression of the dance movement. The human pose detection model used in this project is based on state-of-the-art deep learning techniques, which analyze the positions and movements of different parts of the human body in real-time. This model allows the framework to capture movements of the dancers and translate them into unique visual forms. The case study showcases the potential of "FORMS" by demonstrating how professional young dancers can use the framework to enrich their performance and create new visual perceptions of dance movement. This study contributes to the cultivation of body awareness, understanding of the dance movement and overall enrichment of the art experience. The use of machine learning techniques showcases the potential of technology to enhance and expand the boundaries of artistic expression. The "FORMS" project is a novel and interdisciplinary approach that bridges the fields of art and technology, providing a new way to experience and perceive the dance movement.

CCS CONCEPTS

• **Applied computing** → **Performing arts**; • **Computing methodologies** → **Neural networks**; **Perception**; • **Human-centered computing** → **Empirical studies in interaction design**.

KEYWORDS

Interactive Art, Dance, Machine Learning, Human-Pose Detection

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1 INTRODUCTION

Dance, as an art form, communicates to the audience through the language of body movement [Bartenieff and Lewis 2013; Stevens and McKechnie 2005]. The intersection of dance and computing technology has been explored by professionals from various research fields since the mid-1960s [Herbison-Evans 1991]. Recent advancements in computational art have opened up new possibilities for analyzing and interpreting dance concepts [Craine and Mackrell 2010]. The field of dance has witnessed significant developments in human pose detection, which plays a crucial role in exploring approaches to this art form and materializing movement. The advent of convolutional neural networks (CNNs) and benchmark human pose datasets [Cao et al. 2017; Zhang et al. 2017] has further fueled the intrigue surrounding human pose detection. Leveraging the capabilities of human pose detection, "FORMS" aims to delve deeper into the mechanics of dance movements and generate new visual representations that enrich both the artistic and technical aspects of this art form. By employing machine learning techniques, particularly human pose detection, "FORMS" provides insights into the interconnections of the body during a dance performance. To explore visually the relationship between dance movement and the physical space, we integrated "FORMS" into a live performance. This paper delves into the integration of "FORMS" and presents our findings on how it enhances new perceptions of dance movement. Furthermore, "FORMS" enables dancers to gain a fresh visual understanding of their own movements, expanding their physical vocabulary and fostering creativity in their artistic expression.

2 PROPOSED WORK

Our work revolves around detecting and visually representing human body movement in physical space. The proposed model integrates the OpenPose system [Cao et al. 2017] to identify keypoints on the human body skeleton. Initially, we explored the possibility of connecting these keypoints in various visual forms (Fig. 1). To accomplish this, we conceptualized different representations for visual exploration and subsequently implemented the algorithmic components that enable real-time visual graphics based on the detected poses. In studying the representation of the human skeleton, we sketched different visual forms (Fig. 1) that capture body connections during a dance performance. These representations were categorized into three groups: the relationship between the body

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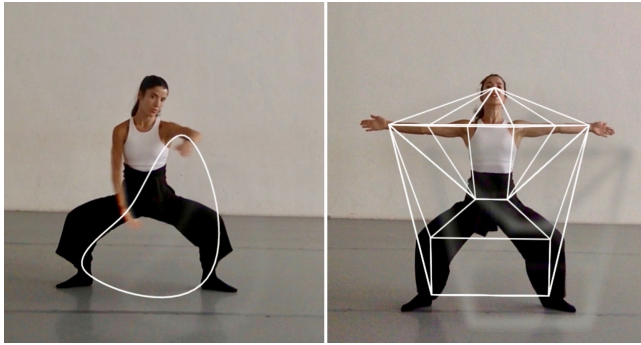


Figure 1: Visual experiments. Left side presents the connection between hands and feet. Right side shows a connection between different keypoints.

extremities (Fig. 1, left side), complex connections between different points (Fig. 1, right side), and a representation of the entire body resembling the human form (Fig. 2). "FORMS" offers numerous possibilities for exhibition, significantly enriching dance movement and providing a fresh visual perspective on the movement. Building upon previous work [Nogueira et al. 2021], we aim to provide personalized and interactive experiences that are highly valued by performers and audiences alike. "FORMS" can be presented in live performances, contributing to a new dimension of the dance movement and engaging both performers and the audience. Involving the public as both performers and spectators, experiencing themselves through "FORMS," is another of our objectives. Our work holds pedagogical potential in dance education, cultivating a mature and organic understanding of movement and promoting greater body awareness from an early age.

3 CASE STUDY

To create a live performance integrating "FORMS," we collaborated with seven young professional dancers who performed in real-time with the technology. The dancers were allowed to select the visual model that best represented their movements from a set of possibilities. The concept underlying the performance revolved around our perception of the world and how often we tend to focus solely on our perspective, disregarding the viewpoints of others. Each visual form and choreographic composition explored this idea, emphasizing the importance of looking at one another from different perspectives. After this point, the interaction between the dancers and "FORMS" was tested for a month, allowing for real-time integration. Following the performance presentation to the public, the dancers were asked to complete a short questionnaire to evaluate their experience. Each dancer provided informed consent, and the survey responses were collected anonymously. The age of young dancers ranged from 17 to 24 years, with an average age of 18.7 years ($SD = 2.56$). In the questionnaire, the dancers were asked to rate their experience on a scale from 0 to 10, with 0 representing a weak experience and 10 indicating an excellent one. The average rating across all participants was 9.7 ($SD = 0.48$). Similarly, the dancers were asked to assess whether they believed the integration of "FORMS" enriched their performance, with an average rating of



Figure 2: Photograph of the performance on stage, where the dancer's movement is detected and enhanced with "FORMS".

9.85 ($SD = 0.37$). In response to an open-ended question regarding whether the perception of their movement was facilitated by the technology, the following statements were obtained: "Observing my movement through the technology made me realize that I need to expand the range of my movements," "Seeing myself from a different perspective helped me to understand the need for clearer and more precise movements, as well as increased focus on the extremities of my body," and "The technology highlighted the potential for more expressive and expansive movements in my arms". Overall, the dancers provided a very positive feedback.

4 CONCLUSION AND FUTURE WORK

The intersection of different fields has led to the creation of "FORMS" and its integration in a real case study. The relationship between dance and machine learning has produced a particularly enriching result for artists and the audience. The outcomes presented have made a positive contribution to this research, and future developments will be built on this foundation. For example, we can experiment visual composition of performances with larger groups. On the other hand, making the experience of this interaction available to the general public can provide gratifying results of well-being and increased body awareness.

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